

Chapter 2

Topological Spaces

In **Lecture 1** we saw that continuity between metric spaces can be characterized purely in terms of open sets (Theorem 3.4 of the Harris handout): f is continuous iff $f^{-1}(\text{open}) = \text{open}$. This means the metric itself is extraneous for the purposes of continuity, limits, and connectedness—all that matters is which sets are declared open. This motivates the following abstraction.

2.1 The Definition

Definition 2.1.1: Topology

A topology on a set X is a collection $\mathcal{T} \subseteq \mathcal{P}(X)$ of subsets of X , called the open sets, satisfying:

- (i) $\emptyset \in \mathcal{T}$ and $X \in \mathcal{T}$.
- (ii) The union of any subcollection of elements of $\mathcal{T}$ is in $\mathcal{T}$.
- (iii) The intersection of any finite subcollection of elements of $\mathcal{T}$ is in $\mathcal{T}$.

A topological space is a pair $(X, \mathcal{T})$. When the topology is clear, we just write X . A subset $U \subseteq X$ is open if $U \in \mathcal{T}$.

Example 2.1.1 (The metric topology)

If (X, d) is a metric space, the collection $\mathcal{T} = \{U \subseteq X : U \text{ is open in the metric sense}\}$ is a topology on X (proved in Lecture 1). This is the metric topology induced by d .

Example 2.1.2 (The discrete and indiscrete topologies)

On any set X : the discrete topology is $\mathcal{T} = \mathcal{P}(X)$ (every subset is open). This is the metric topology for the discrete metric $d(x, y) = 1$ if $x \neq y$. The indiscrete (or trivial) topology is $\mathcal{T} = \{\emptyset, X\}$. In the indiscrete topology, the only open sets are the empty set and X itself.

Example 2.1.3 (The finite complement topology)

Let X be any set. Define $\mathcal{T}_f = \{U \subseteq X : X \setminus U \text{ is finite}\} \cup \{\emptyset\}$. This is a topology: $\emptyset$ and X are in $\mathcal{T}_f$ (since $X \setminus X = \emptyset$ is finite). If $\{U_\alpha\}$ is a collection of elements of $\mathcal{T}_f$, then $X \setminus \bigcup U_\alpha = \bigcap (X \setminus U_\alpha)$, which is an intersection of finite sets, hence finite. If $U_1, \dots, U_n \in \mathcal{T}_f$, then $X \setminus \bigcap U_i = \bigcup (X \setminus U_i)$, a finite union of finite sets, hence finite.

Example 2.1.4 (Topologies on a three-element set)

Let $X = \{a, b, c\}$. Examples of topologies include $\{\emptyset, X\}$ (indiscrete), $\{\emptyset, \{a\}, X\}$, $\{\emptyset, \{a, b\}, \{c\}, X\}$, $\{\emptyset, \{a\}, \{b\}, \{a, b\}, \{b, c\}, X\}$, and $\mathcal{P}(X)$ (discrete). The collection $\{\emptyset, \{a\}, \{b\}, X\}$ is not a topology: it is missing $\{a\} \cup \{b\} = \{a, b\}$.

Note:-

Why bother with topologies that don't come from a metric? In analysis, many natural topologies are not metrizable. The space of bounded functions $f : X \rightarrow \mathbb{R}$ with the topology of pointwise convergence ($f_n \rightarrow f$ iff $f_n(x) \rightarrow f(x)$ for all x) does not come from a metric in general (it is a product topology). The C^∞ topology on smooth functions $\mathbb{R} \rightarrow \mathbb{R}$ is also not metrizable. On the other hand, a metric contains extraneous information: the metrics d, d_1, d_∞ on $\mathbb{R}^n$ all induce the same topology.

2.2 Continuous Maps Between Topological Spaces

Definition 2.2.1: Continuous map

A map $f : X \rightarrow Y$ between topological spaces is continuous if $f^{-1}(U)$ is open in X for every open set $U \subseteq Y$.

Definition 2.2.2: Convergence in topological spaces

A sequence $\{p_n\}$ in a topological space X converges to $p \in X$, written $p_n \rightarrow p$, if for every open set $U \ni p$, there exists N such that $p_n \in U$ for all $n \geq N$. Equivalently, every open set containing p contains all but finitely many p_n .

Note:-

In a general topological space, limits of sequences need not be unique. In the indiscrete topology $\mathcal{T} = \{\emptyset, X\}$, every sequence converges to every point (the only open set containing p is X itself). We will need the Hausdorff condition to guarantee uniqueness.

Proposition 2.2.1 Basic properties of continuous maps

- (a) If $f : X \rightarrow Y$ is continuous and $p_n \rightarrow p$ in X , then $f(p_n) \rightarrow f(p)$ in Y .
- (b) If $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ are continuous, then $g \circ f : X \rightarrow Z$ is continuous.

Proof: (a) Let $V \ni f(p)$ be open in Y . Then $f^{-1}(V)$ is open in X and contains p , so it contains p_n for all but finitely many n . Thus $f(p_n) \in V$ for all but finitely many n .

(b) If $W \subseteq Z$ is open, then $g^{-1}(W)$ is open in Y , so $f^{-1}(g^{-1}(W)) = (g \circ f)^{-1}(W)$ is open in X . $\square$

2.2.1 Comparing Topologies

Definition 2.2.3: Finer and coarser

Given two topologies $\mathcal{T}$ and $\mathcal{T}'$ on X , we say $\mathcal{T}'$ is finer than $\mathcal{T}$ (or $\mathcal{T}$ is coarser than $\mathcal{T}'$) if $\mathcal{T} \subseteq \mathcal{T}'$. If $\mathcal{T} \subsetneq \mathcal{T}'$, we say $\mathcal{T}'$ is strictly finer. Two topologies are comparable if one contains the other.

Note:-

The discrete topology is the finest topology on X (every point is isolated), and the indiscrete topology is the coarsest. A finer topology has more open sets: it is easier for functions into X to be continuous (more

preimages are open), but harder for sequences to converge (more open sets to check). In the discrete topology, a sequence converges iff it is eventually constant. In the indiscrete topology, every sequence converges to every point.

2.3 Bases for a Topology

Keeping track of all open sets is cumbersome. In metric spaces, we started with open balls and generated all open sets as unions. The abstract analogue is a basis.

Definition 2.3.1: Basis for a topology

A basis for a topology on X is a collection $\mathcal{B} \subseteq \mathcal{P}(X)$ satisfying:

- (i) For every $x \in X$, there exists $B \in \mathcal{B}$ with $x \in B$.
- (ii) If $x \in B_1 \cap B_2$ for $B_1, B_2 \in \mathcal{B}$, then there exists $B_3 \in \mathcal{B}$ with $x \in B_3 \subseteq B_1 \cap B_2$.

The topology $\mathcal{T}$ generated by $\mathcal{B}$ consists of all sets that are unions of elements of $\mathcal{B}$. Equivalently, $U \in \mathcal{T}$ iff for every $x \in U$, there exists $B \in \mathcal{B}$ with $x \in B \subseteq U$.

Note:-

Condition (i) ensures $X \in \mathcal{T}$ (since $X = \bigcup_{B \in \mathcal{B}} B$). Condition (ii) ensures finite intersections of open sets are open: if $U_1, U_2 \in \mathcal{T}$ and $x \in U_1 \cap U_2$, pick $B_1, B_2 \in \mathcal{B}$ with $x \in B_i \subseteq U_i$, then (ii) gives $B_3 \in \mathcal{B}$ with $x \in B_3 \subseteq B_1 \cap B_2 \subseteq U_1 \cap U_2$, so $U_1 \cap U_2 \in \mathcal{T}$. Unlike bases in linear algebra, topological bases can contain redundant elements—a better analogy is with generating sets.

Example 2.3.1 (Bases for familiar topologies)

- The open balls $\{B_r(x) : x \in X, r > 0\}$ form a basis for the metric topology on X . Even the rational-radius balls $\{B_{1/n}(x) : x \in X, n \geq 1\}$ suffice.
- The open intervals $\{(a, b) : a < b, a, b \in \mathbb{R}\}$ form a basis for the usual (standard) topology on $\mathbb{R}$. Condition (ii): $(a_1, b_1) \cap (a_2, b_2) = (\max(a_1, a_2), \min(b_1, b_2))$ is either empty or another open interval.

Definition 2.3.2: The lower limit topology

$\mathbb{R}_\ell$ is $\mathbb{R}$ with the topology generated by the basis of half-open intervals $\{[a, b) : a < b\}$. This is strictly finer than the standard topology: every open interval $(a, b) = \bigcup_n [a + 1/n, b)$ is open in $\mathbb{R}_\ell$, but $[0, 1)$ is open in $\mathbb{R}_\ell$ and not in the standard topology.

Definition 2.3.3: The K-topology

$\mathbb{R}_K$ is $\mathbb{R}$ with the topology generated by the basis $\{(a, b) : a < b\} \cup \{(a, b) \setminus K : a < b\}$ where $K = \{1/n : n \in \mathbb{Z}_{>0}\}$. This is strictly finer than the standard topology but is not comparable to $\mathbb{R}_\ell$.

Proposition 2.3.1 Continuity via bases

Let $\mathcal{B}_X$ and $\mathcal{B}_Y$ be bases for topologies on X and Y . Then $f : X \rightarrow Y$ is continuous iff for every $B \in \mathcal{B}_Y$ and every $x \in f^{-1}(B)$, there exists $B' \in \mathcal{B}_X$ with $x \in B' \subseteq f^{-1}(B)$. In metric spaces, this reduces to the familiar ε - δ condition (with $B = B_\varepsilon(f(x))$ and $B' = B_\delta(x)$).

2.4 Subspace, Product, and Quotient Topologies

2.4.1 The Subspace Topology

Definition 2.4.1: Subspace topology

Let $(X, \mathcal{T})$ be a topological space and $Y \subseteq X$. The subspace topology on Y is $\mathcal{T}_Y = \{U \cap Y : U \in \mathcal{T}\}$. That is, a set is open in Y iff it is the intersection of Y with an open set of X .

Note:-

When stating “ U is open,” it is important to specify in which space. The set $(0, 1)$ is open in $\mathbb{R}$, open in $[0, 2]$ (since $(0, 1) = (-1, 1) \cap [0, 2]$), but not open in $\mathbb{R}^2$. The set Y is always open in itself (as $Y = X \cap Y$). If the metric on X restricts to Y , the subspace topology agrees with the induced metric topology: if $\mathcal{T}_X$ comes from a metric d on X , then $\mathcal{T}_Y$ comes from $d|_{Y \times Y}$.

Note:-

The subspace topology is the coarsest topology on Y making the inclusion $\iota : Y \hookrightarrow X$ continuous.

2.4.2 The Product Topology

Definition 2.4.2: Product topology (finite case)

Given topological spaces $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$, the product topology on $X \times Y$ is the topology generated by the basis

$$\mathcal{B} = \{U \times V : U \in \mathcal{T}_X, V \in \mathcal{T}_Y\}.$$

This is a basis: condition (i) holds since $X \times Y \in \mathcal{B}$, and condition (ii) since $(U_1 \times V_1) \cap (U_2 \times V_2) = (U_1 \cap U_2) \times (V_1 \cap V_2) \in \mathcal{B}$.

Note:-

When X, Y are metric spaces, the product topology on $X \times Y$ is the metric topology for $d_\infty((x_1, y_1), (x_2, y_2)) = \max(d_X(x_1, x_2), d_Y(y_1, y_2))$. This agrees with the product topologies for the metrics $d_2 = \sqrt{d_X^2 + d_Y^2}$ and $d_1 = d_X + d_Y$, since all three induce the same topology. The key observation is that $B_r^{d_\infty}((x, y)) = B_r^X(x) \times B_r^Y(y)$, so the metric balls are exactly the basis elements.

Proposition 2.4.1 Continuity into products

A map $f : Z \rightarrow X \times Y$ is continuous iff both components $f_1 = \pi_X \circ f : Z \rightarrow X$ and $f_2 = \pi_Y \circ f : Z \rightarrow Y$ are continuous, where π_X, π_Y are the projection maps. Equivalently, a sequence $(x_n, y_n) \rightarrow (x, y)$ in $X \times Y$ iff $x_n \rightarrow x$ in X and $y_n \rightarrow y$ in Y .

Proof: The projections π_X, π_Y are continuous (preimage of $U \subseteq X$ open is $U \times Y$, which is open). So if f is continuous, then $f_1 = \pi_X \circ f$ and $f_2 = \pi_Y \circ f$ are compositions of continuous maps. Conversely, if f_1, f_2 are continuous and $U \times V$ is a basis element of $X \times Y$, then $f^{-1}(U \times V) = f_1^{-1}(U) \cap f_2^{-1}(V)$, a finite intersection of open sets, hence open. Since every open set in $X \times Y$ is a union of such basis elements, f^{-1} of any open set is open. $\square$

2.4.3 Infinite Products: Box vs. Product Topology

For a family of spaces $\{(X_i, \mathcal{T}_i)\}_{i \in I}$, there are two natural topologies on $\prod_{i \in I} X_i$.

Definition 2.4.3: Box topology

The box topology on $\prod_{i \in I} X_i$ has basis consisting of all products $\prod_{i \in I} U_i$ where each $U_i \subseteq X_i$ is open.

Definition 2.4.4: Product topology

The product topology on $\prod_{i \in I} X_i$ has basis consisting of all products $\prod_{i \in I} U_i$ where each $U_i \subseteq X_i$ is open and $U_i = X_i$ for all but finitely many i .

Note:-

For finite I , the two topologies coincide. For infinite I , the product topology is strictly coarser than the box topology. The product topology is the “right” choice: it is the coarsest topology making all projections $\pi_i : \prod X_j \rightarrow X_i$ continuous. The characterization of continuity extends: $f : Z \rightarrow \prod X_i$ is continuous (in the product topology) iff each component $f_i = \pi_i \circ f$ is continuous.

Example 2.4.1 (The diagonal map)

Let $\Delta : \mathbb{R} \rightarrow \mathbb{R}^{\mathbb{N}}$ be defined by $\Delta(x) = (x, x, x, \dots)$. In the product topology, Δ is continuous (each component $\pi_n \circ \Delta = \text{id}_{\mathbb{R}}$ is continuous). In the box topology, Δ is not continuous: the set $U = \prod_{n=1}^{\infty} (-1/n, 1/n)$ is open in the box topology, but $\Delta^{-1}(U) = \bigcap_{n=1}^{\infty} (-1/n, 1/n) = \{0\}$, which is not open in $\mathbb{R}$.

2.4.4 Homeomorphisms

Definition 2.4.5: Homeomorphism

Two topological spaces X and Y are homeomorphic, written $X \cong Y$, if there exist continuous maps $f : X \rightarrow Y$ and $g : Y \rightarrow X$ with $g \circ f = \text{id}_X$ and $f \circ g = \text{id}_Y$. Equivalently, f is a bijection such that both f and f^{-1} are continuous. Such an f is called a homeomorphism. A homeomorphism is an isomorphism of topological spaces: it induces a bijection $\mathcal{T}_X \leftrightarrow \mathcal{T}_Y$.

Note:-

A continuous bijection need not be a homeomorphism: the inverse may fail to be continuous. If $\mathcal{T}'$ is strictly finer than $\mathcal{T}$ on X , then $\text{id} : (X, \mathcal{T}') \rightarrow (X, \mathcal{T})$ is a continuous bijection (since $\mathcal{T} \subseteq \mathcal{T}'$), but $\text{id}^{-1} : (X, \mathcal{T}) \rightarrow (X, \mathcal{T}')$ is not continuous (some $U \in \mathcal{T}' \setminus \mathcal{T}$ fails to pull back to an open set).

Example 2.4.2

$f : (-\pi/2, \pi/2) \rightarrow \mathbb{R}$ defined by $f(x) = \tan(x)$ is a homeomorphism (both $\tan$ and $\arctan$ are continuous). So $\mathbb{R}$ is homeomorphic to any bounded open interval. This shows boundedness is not a topological property.

2.5 Closed Sets, Closure, Interior, Boundary

All the definitions from Lecture 1 carry over verbatim to general topological spaces.

Definition 2.5.1: Closed set

A subset A of a topological space X is closed if $X \setminus A$ is open. By De Morgan: $\emptyset$ and X are closed, arbitrary intersections of closed sets are closed, and finite unions of closed sets are closed.

Note:-

A set can be both open and closed (clopen): $\emptyset$ and X always are, and in the discrete topology every set is clopen. A set can also be neither open nor closed: $[0, 1]$ in the standard topology on $\mathbb{R}$, or $\mathbb{Q} \subset \mathbb{R}$.

Definition 2.5.2: Closure, interior, boundary

Let $A \subseteq X$ be any subset.

- The closure of A is $\bar{A} = \bigcap \{F \supseteq A : F \text{ closed}\}$, the smallest closed set containing A . A point $x \in \bar{A}$ iff every open set containing x meets A .
- The interior of A is $\text{Int}(A) = \bigcup \{U \subseteq A : U \text{ open}\}$, the largest open set contained in A . A point $x \in \text{Int}(A)$ iff some open set containing x is contained in A .
- The boundary of A is $\partial A = \bar{A} \setminus \text{Int}(A)$. A point $x \in \partial A$ iff every open set containing x meets both $\bar{A}$ and $X \setminus A$.

Note:-

The complementary identities hold: $\overline{X \setminus A} = X \setminus \text{Int}(A)$ and $\text{Int}(X \setminus A) = X \setminus \bar{A}$. Also: A is closed iff $\bar{A} = A$, and A is open iff $\text{Int}(A) = A$.

Example 2.5.1

In $\mathbb{R}$ with the standard topology: if $A = [0, 1)$, then $\bar{A} = [0, 1]$, $\text{Int}(A) = (0, 1)$, and $\partial A = \{0, 1\}$.

Definition 2.5.3: Neighborhood

An open set U containing a point p is called a neighborhood of p . Then: $p \in \text{Int}(A)$ iff A contains a neighborhood of p , and $p \in \bar{A}$ iff every neighborhood of p intersects A .

Definition 2.5.4: Dense subset

A is dense in X if $\bar{A} = X$, i.e., every nonempty open set intersects A . For instance, $\mathbb{Q}$ is dense in $\mathbb{R}$ (standard topology).

2.6 Limit Points and the Hausdorff Condition

2.6.1 Limit Points

Definition 2.6.1: Limit point

A point $x \in X$ is a limit point of $A \subseteq X$ if every neighborhood U of x satisfies $U \cap (A \setminus \{x\}) \neq \emptyset$. The set of limit points of A is denoted A' . A point of A that is not a limit point is an isolated point of A .

Proposition 2.6.1 Closure via limit points

$\bar{A} = A \cup A'$. In particular, A is closed iff $A' \subseteq A$ (iff A contains all its limit points).

Proof: If $x \in A$, every neighborhood meets A (it contains x). If $x \in A'$, every neighborhood meets $A \setminus \{x\} \subseteq A$. Either way $x \in \bar{A}$. Conversely, if $x \notin A$ and $x \notin A'$, then some neighborhood U satisfies $U \cap (A \setminus \{x\}) = \emptyset$; since $x \notin A$, this gives $U \cap A = \emptyset$, so $x \notin \bar{A}$. $\square$

Example 2.6.1

In $\mathbb{R}_{\text{std}}$: 1 is a limit point of both $(0, 1)$ and $[0, 1]$. In the set $S = \{1/n : n \geq 1\} \cup \{0\}$, the point 0 is a limit point (every neighborhood of 0 contains $1/n$ for large n), but 1 is an isolated point (the ball $B_{1/2}(1) \cap S = \{1\}$).

Note:-

In metric spaces, $x \in \overline{A}$ iff there exists a sequence in A converging to x , and x is a limit point of A iff there exists a sequence in $A \setminus \{x\}$ converging to x . In general topological spaces, sequences are insufficient: the correct tool is nets or filters. However, in spaces with a countable basis of neighborhoods at each point (“first-countable” spaces, which include all metric spaces), sequences suffice.

2.6.2 Hausdorff Spaces

In the indiscrete topology on $\mathbb{R}$, every sequence converges to every point, and in the finite complement topology on $\mathbb{R}$, a sequence of distinct points converges to every point of $\mathbb{R}$. To exclude such pathologies:

Definition 2.6.2: Hausdorff space

A topological space X is Hausdorff (or T_2) if for every pair of distinct points $x_1 \neq x_2 \in X$, there exist disjoint open sets U_1, U_2 with $x_1 \in U_1$ and $x_2 \in U_2$.

Theorem 2.6.1 Hausdorff implies unique limits

If X is Hausdorff, then every sequence in X converges to at most one limit.

Proof: Suppose $x_n \rightarrow x$ and $y \neq x$. Choose disjoint open sets $U_x \ni x$ and $U_y \ni y$. Since $x_n \rightarrow x$, there exists N with $x_n \in U_x$ for all $n \geq N$. But then $x_n \notin U_y$ for $n \geq N$ (since $U_x \cap U_y = \emptyset$), so $\{x_n\}$ does not converge to y . $\square$

Example 2.6.2 (Hausdorff and non-Hausdorff spaces)

- (a) Every metric space is Hausdorff: if $x \neq y$, set $r = d(x, y)/2 > 0$ and take $U = B_r(x)$, $V = B_r(y)$.
- (b) The finite complement topology on $\mathbb{R}$ is not Hausdorff: any two nonempty open sets have finite complements, so their intersection is $\mathbb{R}$ minus a finite set, hence nonempty. But it is T_1 (singletons are closed, since $\mathbb{R} \setminus \{x\}$ has finite complement $\{x\}$).
- (c) The discrete topology is always Hausdorff ($U_i = \{x_i\}$).
- (d) Subspaces of Hausdorff spaces are Hausdorff. Products of Hausdorff spaces are Hausdorff.

Note:-

There is a hierarchy of separation axioms: T_1 (points are closed) $\Leftarrow T_2$ (Hausdorff) $\Leftarrow T_3$ (regular) $\Leftarrow T_4$ (normal). The finite complement topology on $\mathbb{R}$ is T_1 but not T_2 . We will primarily work with Hausdorff spaces. Every metric space is T_4 (hence satisfies all separation axioms).

2.7 Connectedness

Definition 2.7.1: Connected space

A topological space X is connected if it cannot be written as $X = U \cup V$ where U, V are disjoint, nonempty, open sets. Such a decomposition is called a separation of X . A subset $A \subseteq X$ is connected if it is connected in the subspace topology.

Proposition 2.7.1 $[0, 1]$ is connected

The interval $[0, 1] \subset \mathbb{R}$ (with the standard topology) is connected.

Proof: Suppose $[0, 1] = U \cup V$ is a separation with $0 \in U$. Let $a = \sup\{x \in [0, 1] : [0, x] \subseteq U\}$. Since U is open and $0 \in U$, we have $a > 0$.

If $a \in V$: since V is open in $[0, 1]$, there exists $\varepsilon > 0$ with $(a - \varepsilon, a] \cap [0, 1] \subseteq V$. But by definition of a , the interval $[0, a - \varepsilon)$ is not entirely in U , so some point of $[0, a - \varepsilon)$ is in V . But then there exist points of V below $a - \varepsilon$, and $[0, x] \subseteq U$ fails before $a - \varepsilon$, contradicting $a = \sup$.

More carefully: if $a < 1$ and $a \in U$, then U is open, so $(a - \varepsilon, a + \varepsilon) \subseteq U$ for some ε , contradicting $a = \sup$. If $a \in V$, then $(a - \varepsilon, a + \varepsilon) \cap [0, 1] \subseteq V$, so $[0, x] \subseteq U$ for $x < a - \varepsilon$ only, giving $\sup \leq a - \varepsilon < a$, contradiction. So $a = 1$, meaning $U = [0, 1]$ and $V = \emptyset$. $\square$

Theorem 2.7.1 Continuous image of a connected space is connected

If $f : X \rightarrow Y$ is continuous and X is connected, then $f(X) \subseteq Y$ is connected.

Proof: Suppose $f(X) = U \cup V$ is a separation (in the subspace topology). Then $f^{-1}(U)$ and $f^{-1}(V)$ are disjoint, nonempty (since $U, V \subseteq f(X)$), and open in X (since f is continuous and U, V are open in $f(X)$, which means $U = U' \cap f(X)$ for some open $U' \subseteq Y$, so $f^{-1}(U) = f^{-1}(U')$ is open). Their union is X , giving a separation of X . Contradiction. $\square$

Corollary 2.7.1 Intermediate Value Theorem

Let X be a connected topological space and $f : X \rightarrow \mathbb{R}$ continuous. If $a, b \in X$ and r lies between $f(a)$ and $f(b)$, then $r \in f(X)$: there exists $c \in X$ with $f(c) = r$.

Proof: The image $f(X)$ is a connected subset of $\mathbb{R}$. If $r \notin f(X)$, then $U = f(X) \cap (-\infty, r)$ and $V = f(X) \cap (r, \infty)$ give a separation of $f(X)$ (both nonempty since $f(a)$ is on one side and $f(b)$ on the other). Contradiction. $\square$

Theorem 2.7.2 Connected subsets of $\mathbb{R}$ are intervals

A subset $A \subseteq \mathbb{R}$ is connected (in the standard topology) if and only if A is an interval (i.e., $x, z \in A$ and $x < y < z$ implies $y \in A$).

2.7.1 Path-Connectedness

Definition 2.7.2: Path

A path in X from x to y is a continuous map $\gamma : [0, 1] \rightarrow X$ with $\gamma(0) = x$ and $\gamma(1) = y$. (Here $[0, 1]$ has the subspace topology from $\mathbb{R}$.)

Definition 2.7.3: Path-connected

A space X is path-connected if for every pair $x, y \in X$, there exists a path from x to y . The relation $x \sim y$ (“connected by a path”) is an equivalence relation; the equivalence classes are the path components of X .

Theorem 2.7.3 Path-connected implies connected

If X is path-connected, then X is connected.

Proof: Suppose $X = U \cup V$ is a separation. Pick $x \in U, y \in V$, and let $\gamma : [0, 1] \rightarrow X$ be a path from x to y . Then $\gamma^{-1}(U)$ and $\gamma^{-1}(V)$ are disjoint open sets covering $[0, 1]$, with $0 \in \gamma^{-1}(U)$ and $1 \in \gamma^{-1}(V)$, giving a separation of $[0, 1]$. This contradicts the connectedness of $[0, 1]$. $\square$

Note:-

The converse is false in general: the topologist’s sine curve $S = \{(x, \sin(1/x)) : x > 0\} \cup \{(0, y) : -1 \leq y \leq 1\}$ is connected but not path-connected. However, for open subsets of $\mathbb{R}^n$, connected and path-connected are equivalent.

Theorem 2.7.4 Products of connected spaces are connected

If X_i is connected for each $i \in I$, then $\prod_{i \in I} X_i$ (with the product topology) is connected.

2.8 Exercises

Exercise 2.8.1 Topology exercises

- (1) Verify that the finite complement topology $\mathcal{T}_f$ on an infinite set X is T_1 but not Hausdorff.
- (2) Let X be a set with topologies $\mathcal{T}$ and $\mathcal{T}'$, with $\mathcal{T}'$ strictly finer than $\mathcal{T}$. Show that $\text{id} : (X, \mathcal{T}') \rightarrow (X, \mathcal{T})$ is continuous but not a homeomorphism.
- (3) Show that if X is Hausdorff, then $Y \subseteq X$ (with the subspace topology) is Hausdorff, and $X \times Y$ (with the product topology) is Hausdorff whenever both factors are.
- (4) Prove that $A \subseteq \mathbb{R}$ (standard topology) is connected iff A is an interval.
- (5) Let X be a topological space and $\{A_i\}_{i \in I}$ a family of connected subspaces with $\bigcap_{i \in I} A_i \neq \emptyset$. Show that $\bigcup_{i \in I} A_i$ is connected.
- (6) Show that the closure of a connected set is connected.
- (7) Prove that a finite topological space is metrizable if and only if the topology is discrete.

Solution

- (1) T_1 : for any $x \in X$, $X \setminus \{x\}$ has complement $\{x\}$, which is finite, so $X \setminus \{x\} \in \mathcal{T}_f$. Thus singletons are closed. Not Hausdorff: if U, V are nonempty open sets, then $X \setminus U$ and $X \setminus V$ are finite, so $X \setminus (U \cap V) = (X \setminus U) \cup (X \setminus V)$ is finite. Since X is infinite, $U \cap V$ is infinite (in particular nonempty). So no two nonempty open sets are disjoint.
- (2) Continuity: if $U \in \mathcal{T}$, then $U \in \mathcal{T}'$ (since $\mathcal{T} \subseteq \mathcal{T}'$), so $\text{id}^{-1}(U) = U$ is open in $\mathcal{T}'$. Not a homeomorphism: since $\mathcal{T}' \supsetneq \mathcal{T}$, there exists $V \in \mathcal{T}' \setminus \mathcal{T}$. Then $(\text{id}^{-1})^{-1}(V) = V \notin \mathcal{T}$, so id^{-1} is not continuous.

(3) Subspace: if $y_1 \neq y_2 \in Y \subseteq X$, take disjoint open $U_1, U_2 \subseteq X$ separating them. Then $U_1 \cap Y, U_2 \cap Y$ are disjoint open sets in Y separating y_1, y_2 . Product: given $(x_1, y_1) \neq (x_2, y_2)$ in $X \times Y$, WLOG $x_1 \neq x_2$. Choose disjoint open $U_1, U_2 \subseteq X$ with $x_i \in U_i$. Then $U_1 \times Y$ and $U_2 \times Y$ are disjoint open sets in $X \times Y$ separating the two points.

(4) ($\Rightarrow$) If A is not an interval, there exist $x < y < z$ with $x, z \in A$ and $y \notin A$. Then $A = (A \cap (-\infty, y)) \cup (A \cap (y, \infty))$ is a separation. ($\Leftarrow$) If A is an interval and $A = U \cup V$ is a separation with $u \in U, v \in V$, WLOG $u < v$. Let $a = \sup(U \cap [u, v])$. If $a \in U$: U open gives $(a - \varepsilon, a + \varepsilon) \cap A \subseteq U$, contradicting $a = \sup$ (there are points of U beyond a). If $a \in V$: V open gives $(a - \varepsilon, a + \varepsilon) \cap A \subseteq V$, so $\sup(U \cap [u, v]) \leq a - \varepsilon < a$, contradiction.

(5) Suppose $Y = \bigcup A_i = U \cup V$ is a separation. Pick $p \in \bigcap A_i$. WLOG $p \in U$. For each i , the sets $U \cap A_i$ and $V \cap A_i$ are disjoint open sets in A_i covering A_i , with $p \in U \cap A_i \neq \emptyset$. Since A_i is connected, $V \cap A_i = \emptyset$ for all i . Thus $V \cap Y = \bigcup (V \cap A_i) = \emptyset$, contradicting $V \neq \emptyset$.

(6) Let A be connected and suppose $\bar{A} = U \cup V$ is a separation. Then $A = (U \cap A) \cup (V \cap A)$ with $U \cap A$ and $V \cap A$ open in A and disjoint. Since A is connected, one is empty, say $V \cap A = \emptyset$, so $A \subseteq U$. But then $\bar{A} \subseteq \bar{U}$. Since U is open in $\bar{A}$ and V is its complement in $\bar{A}$, U is also closed in $\bar{A}$, giving $\bar{U} \cap \bar{A} = U$. So $\bar{A} \subseteq U$, meaning $V = \bar{A} \setminus U = \emptyset$, contradiction.

(7) ($\Leftarrow$) The discrete topology on a finite set $X = \{x_1, \dots, x_n\}$ is metrizable (use the discrete metric). ($\Rightarrow$) Suppose X is finite and metrizable with metric d . For each $x \in X$, let $r_x = \min\{d(x, y) : y \in X, y \neq x\} > 0$ (the minimum exists since X is finite). Then $B_{r_x}(x) = \{x\}$, so singletons are open. Since every subset is a union of singletons, every subset is open, and the topology is discrete.